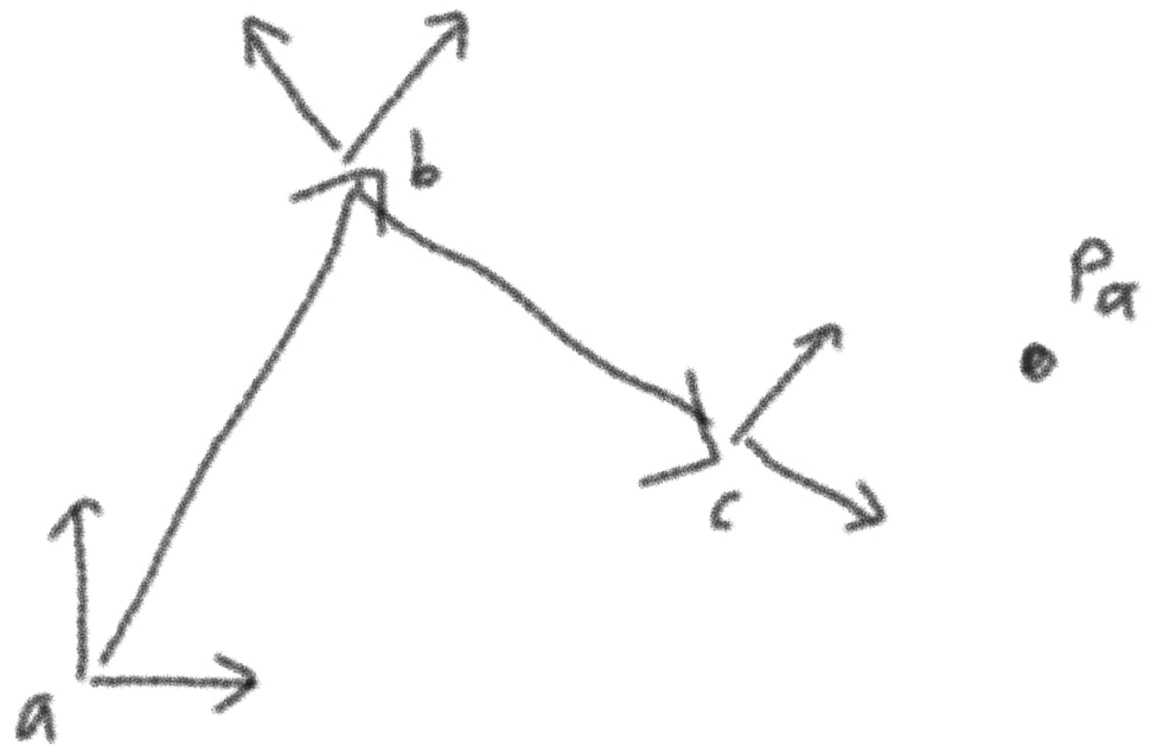
4D/6D

1st exam:

1. 2D transformation matrices

   Given 3 coordinate systems and a point p:
   i) Write the transformation matrices Ta->b, Tb->c as the product of rotation and translation matrices.
   ii) Calculate the point p in coordinates c.

2. write the intrinsic matrix K and unprojection formula.

3. 2D Marching Cubes: given the grid and sdf values draw the vertices and edges, with isosurface level = 0.

4. TSDF Online formula
   i) Write the formula
   ii) Given the initial values and tsdf and w values at each step compute 2 steps.

2nd exam:

1. Skinned Multi-Person Linear Model
   i) explain each parameter in the skinning model - T, J, w, theta.
   ii) which parameters are dependent on theta and beta.
   iii) write down linear blend skinning formula.

2. rendering function - write and explain each term

3. f(x) = (x-3)^2
   i) write down an optimization step formula.

ii) calculate x1 and x2.
iii) what if tk step is too large? too small?

4. MLP
   Given a MLP graph:
   i) write down parameterized function.
   ii) what are W and b dimensions?
   iii) what does d/||d|| means geometrically apart from gradient ascent direction?

3rd exam:

1. Radiative Transfer Equation
   i) Briefly explain all the terms.
   ii) what assumptions are made to derive the NeRF formula? (piecewise constant color and density)
   iii) how the scene is reconstructed from different camera views? (triangulation)
   iv) write down the gamma function and explain why is it needed.

2. Densification
   Complete the drawings:

- Under-reconstruction (clone)
- Over-reconstruction (split)

3. Inverse Kinematics
   theta = f^(-1)(t)
   i) explain why it is not trivial to calculate the inverse of f.
   ii) how to approximate it? (least squares formula)

4. given the list of motion tasks:

- Motion Adaptation / Character Control
- Motion Retargeting
- Motion Completion
- Motion Creation

describe them, state what is available and what is the result.